

Shared Nominal Models for Fleet-Wide Condition Monitoring

Independent Operational-Radius Convergence Across Heterogeneous Assets

Eduardo Di Santi¹

¹Department of Applied Mathematics, University of Colorado Boulder

Abstract

Transportation operators increasingly face the challenge of deploying condition-monitoring systems across large fleets of heterogeneous assets without fault labels, asset-specific retraining, or lengthy calibration campaigns. We propose a day-one deployment method that explicitly separates what is shared across a fleet from what must remain local to each operational asset. A single nominal representation is trained once, offline, using only healthy observations, and is subsequently deployed unchanged across all assets. At deployment, each asset independently estimates its own operational radius from a short stream of unlabeled nominal observations while preserving the shared representation. We hypothesize that, under stable operation, this radius converges rapidly to an asset-specific value; after convergence, persistent departures from the converged radius are interpreted as evidence of degradation rather than absorbed through continued adaptation. The proposed deployment strategy is representation-agnostic and does not rely on fault labels or per-asset retraining. It is validated on public rolling-element bearing datasets, where it preserves 98–100% fault detection while demonstrating that transferring an absolute threshold from simulation fails catastrophically, whereas transferring only a standardized calibration anchor and re-expressing it in each asset’s local statistics recovers near-native performance. The resulting deployment primitive consists of a shared nominal representation, per-asset operational-radius convergence, and lightweight edge inference. A compact theoretical rationale is provided in the appendix as an explanatory justification rather than as a prerequisite for deployment.

Keywords: condition monitoring, condition-based maintenance, transportation asset management, fleet deployment, anomaly detection, cold start, local calibration, edge monitoring

1 Introduction

Modern transportation systems increasingly rely on condition-monitoring technologies to supervise heterogeneous operational assets, including rolling-element bearings, axleboxes, traction motors, gearboxes, pumps, actuators, and other components whose condition can be inferred from repeated observations. Despite the rapid progress of machine learning for anomaly detection, deploying a monitoring system across an operational fleet remains fundamentally a deployment problem rather than a representation-learning problem.

At fleet rollout, four constraints typically occur simultaneously. First, the target fleet contains few or no labeled fault examples. Second, waiting months or years for representative failures is operationally unacceptable. Third, installation-specific factors such as sensor coupling, mounting, operating conditions, and environmental noise modify the numerical scale of anomaly scores from one asset to another. Fourth, maintaining a separate model or manually calibrated threshold for every operational unit does not scale operationally.

The practical question is therefore not simply

How can a fault be detected?

but rather

How can a shared condition-monitoring model be deployed on day one across many heterogeneous assets without fault labels, per-asset retraining, or manual threshold tuning?

We address this deployment problem by separating the diagnostic representation from the local decision boundary. The representation is learned once, offline, and shared across the fleet. Calibration remains local: each operational asset estimates its own operational radius from a short warm-up of unlabeled nominal observations while preserving the common representation. Under stable operation, the proposed calibration process is expected to drive each asset toward its own operational radius,

$$\tau_k(t) \rightarrow \tau_k^*,$$

without modifying the shared model. This convergence hypothesis is evaluated experimentally in Section 5.

Once convergence has been reached, the radius is frozen or tightly bounded so that progressive degradation cannot redefine itself as nominal.

This distinction is motivated by a practical observation. Simulation-to-field transfer frequently fails because calibration does not transfer, even when the learned representation remains valid. Absolute anomaly thresholds depend on installation-specific score scales, whereas standardized statistical anchors can be transferred and subsequently re-expressed using local residual statistics. The deployment procedure proposed here therefore transfers only the calibration rule, not the absolute decision boundary.

The contribution of this paper is consequently a deployment methodology rather than a new neural-network architecture. Although the experimental validation uses a lightweight Conv1D autoencoder as the shared nominal representation, the deployment mechanism itself is independent of the underlying representation model. The principal contributions are:

1. a day-one deployment architecture based on one shared nominal representation and independently calibrated operational assets;
2. a standardized simulator- or validation-derived calibration anchor that initializes local decision boundaries without transferring absolute score scales;
3. an operational-radius convergence criterion that separates cold-start calibration from subsequent degradation monitoring;
4. a guarded adaptation mechanism that prevents anomalous observations from inflating their own acceptance boundary; and
5. an experimental protocol that evaluates cold-start behaviour, convergence, simulation-to-field transfer, and post-convergence anomaly detection.

The methodology is intentionally asset-agnostic. It requires only a shared nominal representation together with repeated observations from independently monitored operational units. In this paper, we validate the approach using public rolling-element bearing datasets (CWRU and NASA IMS), which provide controlled nominal, fault, and run-to-failure conditions. Bearings should therefore be interpreted as the first experimental validation domain rather than as a limitation of the proposed deployment methodology.

The remainder of the paper is organized as follows. Section 2 reviews condition monitoring, one-class anomaly detection, adaptive calibration, and fleet deployment. Section 3 presents the proposed deployment methodology. Section 4 describes the experimental protocol. Section 5 reports the deployment-oriented results, while Section 6 discusses implications, limitations, and future extensions. A compact theoretical justification is provided in Appendix A.

2 Related Work

Condition-based maintenance (CBM) has become a central component of modern transportation asset management, with condition monitoring providing the operational evidence required to supervise large fleets of heterogeneous assets [10, 14]. Although the sensing modality depends on the application—including vibration, acoustic emission, electrical current, temperature, strain, or other physical measurements—the computational objective is fundamentally the same: to characterize nominal operation and detect persistent departures that may indicate degradation.

Most machine-learning approaches formulate this problem as supervised fault classification, assuming representative fault examples are available during training. While effective in benchmark datasets, supervised methods require fault labels and often assume that deployment conditions closely match those observed during model development.

One-class and novelty-detection methods, including one-class SVM [4], neural-network novelty detection [5], and autoencoder-based anomaly detection [7, 8], reduce the dependence on fault labels by learning only nominal behaviour. Existing work primarily studies how to learn anomaly detectors, whereas comparatively little attention has been paid to the deployment problem itself: how a single nominal model should be initialized, calibrated, and maintained across a heterogeneous fleet of independently operating assets.

Adaptive thresholds are frequently employed in streaming anomaly detection to compensate for changes in operating conditions. However, unrestricted adaptation introduces an important operational risk: persistent degradation may progressively redefine itself as nominal by continuously shifting the decision boundary. This observation motivates separating calibration from monitoring. In the proposed methodology, adaptation is restricted to a finite cold-start phase whose objective is to estimate an operational radius for each asset. Once convergence has been achieved, the shared representation remains fixed and subsequent departures become diagnostic evidence rather than input for further adaptation.

The present work is also related to invariant representation learning [9] and anomaly detection under distribution shift [19], which seek representations that are insensitive to operationally irrelevant variability. Our contribution lies at a different layer of the condition-monitoring pipeline. Rather than proposing a new representation-learning architecture, we address the operational deployment problem itself: how a shared nominal representation can be deployed across heterogeneous assets through standardized initialization, independent operational-radius convergence, and local calibration, without fault labels or per-asset retraining.

The methodology is intentionally independent of both sensing modality and asset type. Public rolling-element bearing datasets (CWRU and NASA IMS) are used in this paper solely as controlled experimental benchmarks because they provide nominal operation, multiple fault modes, and run-to-failure data under well-characterized conditions.

Accordingly, this paper should be interpreted not as proposing another anomaly-detection architecture, but as introducing a deployment methodology in which a shared nominal representation is combined with independent operational-radius convergence for each monitored asset.

3 Method

3.1 Deployment architecture: shared representation, local radius

Let $x_{k,t} \in \mathcal{X}$ denote an observation window acquired from operational unit k at time t . In the bearing validation presented in this paper, $x_{k,t}$ is instantiated as a fixed-length vibration window. A shared offline-trained representation model produces an anomaly score

$$r_{k,t} = r(x_{k,t}). \quad (1)$$

The representation model is common to the fleet. The decision boundary is not. Each asset maintains a local operational radius $\tau_k(t)$ and raises an alarm when

$$r_{k,t} > \tau_k(t). \quad (2)$$

The method therefore enforces three deployment rules:

1. the representation is shared;
2. the calibration is local;
3. the model is never retrained per asset.

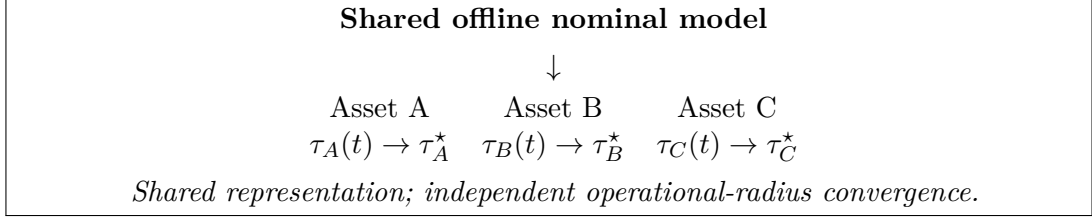


Figure 1: Fleet deployment architecture. One representation is distributed fleet-wide, while each physical asset converges independently to its own operational radius.

Definition 1 (Operational radius). *For a fixed nominal representation and an independently monitored operational unit k , the operational radius τ_k^* is the converged, asset-specific acceptance boundary estimated from nominal deployment observations. It calibrates the anomaly decision locally without modifying or retraining the shared representation.*

3.2 Representation instantiation

The deployment method is independent of the representation backbone. In this paper, we instantiate it with a Conv1D autoencoder $A_\phi = D_\phi \circ E_\phi$ trained exclusively on nominal signals. The encoder maps a 1200-sample window through three Conv1D layers (32, 64, and 128 filters) to a 16-dimensional latent code with an ℓ_1 activity penalty; the decoder mirrors this structure with Conv1DTranspose layers. The model has 847,601 parameters and is trained with Adam, mean-squared reconstruction loss, early stopping, and learning-rate decay.

The anomaly score is the reconstruction residual

$$r(x) = \|x - A_\phi(x)\|_2^2. \quad (3)$$

The autoencoder is an implementation choice, not the scientific contribution. Any nominal representation producing a stable scalar distance or anomaly score could replace Equation 3.

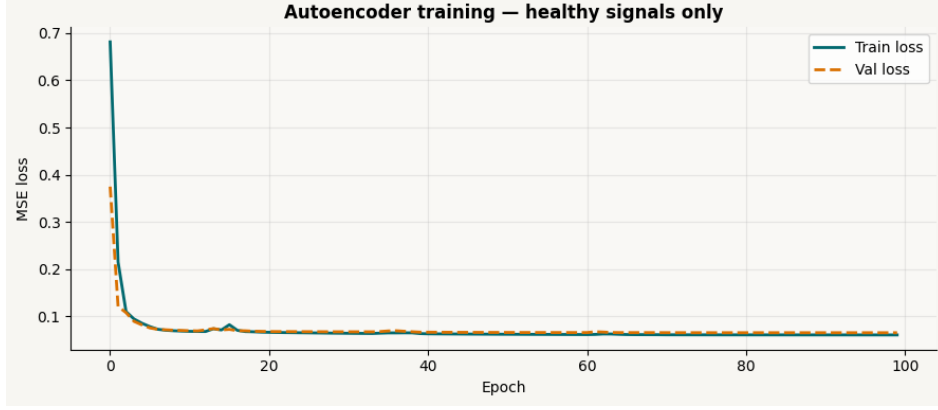


Figure 2: Training and validation loss of the healthy-only Conv1D autoencoder used as the shared nominal representation.

3.3 Standardized anchor for day-one initialization

From held-out nominal validation data or a nominal simulator, we estimate the residual mean μ_0 and standard deviation σ_0 , choose a target nominal percentile, and compute the standardized anchor

$$z_0 = \frac{\tau_0 - \mu_0}{\sigma_0}. \quad (4)$$

The absolute value τ_0 is not transferred to deployment. Only the dimensionless anchor z_0 is shared across the fleet.

For a newly deployed asset k , the local threshold is

$$\tau_k(t) = \mu_k(t) + z_0 \sigma_k(t), \quad (5)$$

where $\mu_k(t)$ and $\sigma_k(t)$ are estimated from the asset’s accepted nominal residuals.

This construction separates the shape of the fleet-wide decision rule from the local residual scale induced by mounting, sensor gain, load, and environmental noise.

3.4 Operational-radius convergence

During cold start, each asset collects a short sequence of accepted nominal residuals and updates Equation 5. The asset is considered calibrated when the radius stabilizes.

For a look-back interval L , define the convergence statistic

$$C_k(t) = \frac{|\tau_k(t) - \tau_k(t - L)|}{\max(\tau_k(t - L), \epsilon)}, \quad (6)$$

where $\epsilon > 0$ prevents division by zero.

Calibration is declared when

$$C_k(t) < \epsilon_C \quad (7)$$

for m consecutive updates. The converged operational radius is then

$$\tau_k^* = \tau_k(t_{\text{conv}}). \quad (8)$$

The key hypothesis evaluated in this paper is:

Under stable nominal operation, every asset converges rapidly to its own operational radius around a fixed shared representation.

After convergence, τ_k^* is frozen or allowed only a tightly bounded robust correction. This prevents a slowly increasing anomaly score from being absorbed as a new nominal state.

3.5 Guarded updates

Only confidently nominal samples are admitted to the local baseline. Let $\delta > 0$ define a guard band. A residual updates the local statistics only when

$$r_{k,t} < \tau_k(t) - \delta. \quad (9)$$

Samples inside the guard band are observed but do not update the baseline, and samples above $\tau_k(t)$ are flagged as anomalous. The three regions are therefore:

$$r_{k,t} < \tau_k(t) - \delta \quad \Rightarrow \quad \text{safe update,} \quad (10)$$

$$\tau_k(t) - \delta \leq r_{k,t} \leq \tau_k(t) \quad \Rightarrow \quad \text{observe, do not update,} \quad (11)$$

$$r_{k,t} > \tau_k(t) \quad \Rightarrow \quad \text{anomaly.} \quad (12)$$

This rule is deliberately lightweight: each asset stores only a residual buffer or running robust statistics, its current radius, and its convergence state.

3.6 Deployment states

Each asset transitions through three operational states:

1. **Cold start:** the standardized anchor is re-expressed in local statistics;
2. **Converged nominal:** τ_k^* has stabilized and monitoring is active;
3. **Departure:** persistent scores exceed the converged radius.

A practical alarm may require q exceedances within the last p observations to avoid reacting to isolated impulses.

3.7 Scope of the Method

The deployment mechanism proposed in this paper is intentionally asset-agnostic. Throughout the paper, the index k denotes an independently monitored operational unit. Depending on the application, k may represent a rolling-element bearing, a traction motor, a gearbox, a pump, an actuator, a vehicle, or a spatially indexed infrastructure segment.

The methodology requires only: (i) a shared nominal representation, (ii) repeated observations from each operational unit, and (iii) local estimation of an operational radius.

The present manuscript validates the approach on public bearing datasets because they provide controlled nominal, fault, and run-to-failure conditions. Bearings should therefore be interpreted as the first validation domain rather than as a limitation of the proposed deployment method.

4 Experimental Design

The experiments are designed around the deployment claim, not around the autoencoder architecture. Every stage therefore measures some subset of the following quantities:

- false-alarm rate during cold start;
- number of accepted samples to operational-radius convergence;
- variability of τ_k^* across assets and regimes;

- post-convergence detection rate;
- sensitivity to buffer size and convergence parameters;
- performance of absolute-threshold transfer versus standardized-anchor transfer;
- cost of per-asset retraining avoided by the shared-model design.

Stage 1 — Controlled multi-asset simulation. Four bearing streams are generated with a physics-motivated nominal simulator using 1200-sample windows, a 30 Hz shaft frequency, and bounded variation in amplitude, phase, speed, offset, and noise. The shared model is trained on 800 nominal windows, with 200 held out to derive z_0 . Each simulated asset begins in nominal operation and independently calibrates its radius. Asset A remains nominal; Assets B–D transition after convergence to outer-race, inner-race, and ball-fault regimes.

This stage measures: (i) samples to convergence; (ii) false alarms before and after convergence; (iii) the separation between $r_{k,t}$ and τ_k^* after fault onset; and (iv) whether each nominal asset reaches a stable but asset-specific radius.

Stage 2 — Direct simulation-to-field deployment on CWRU. The nominal representation trained in Stage 1 exclusively on synthetic healthy signals is transferred unchanged to the Case Western Reserve University (CWRU) Drive-End data. No CWRU signal—healthy or faulty—is used to retrain, fine-tune, or otherwise modify the representation model. The purpose of this stage is to separate representation transfer from decision-boundary transfer.

Each CWRU operating condition is treated as an independent deployment environment. A short stream of unlabeled healthy observations is used only to estimate the local residual statistics $\mu_k(t)$ and $\sigma_k(t)$ and, consequently, the asset-specific operational radius $\tau_k(t)$. Fault observations are then evaluated using the same simulator-trained model and the locally calibrated decision boundary.

This stage tests two distinct hypotheses: (i) whether the nominal representation learned from simulation transfers directly to real field signals without retraining; and (ii) whether the standardized calibration anchor transfers even though the absolute simulator threshold does not.

The main outcome is not merely fault classification. It is whether each stream reaches a stable τ_k^* during warm-up and then detects the subsequent departure without model retraining.

Stage 3a — Baselines. Throughout this ablation, the simulator-trained representation model is held fixed. Only the decision-boundary calibration strategy is changed. We compare the autoencoder residual against: (i) one-class SVM on raw windows; (ii) one-class SVM on latent codes; (iii) elliptical envelope on raw features; and (iv) elliptical envelope on latent codes. Every detector uses the same standardized-anchor and local-radius procedure so that differences are attributable to the representation and score rather than to unequal calibration.

Table 1: [PLACEHOLDER] Detection and cold-start performance under a common local-radius calibration procedure.

Method	Healthy FAR	Ball DR	Inner-race DR	Outer-race DR	Median samples to convergence	Per-asset retraining
AE residual (ours)	11.8	99.0	100.0	100.0	TODO	No
OC-SVM (raw)	TODO	TODO	TODO	TODO	TODO	No
OC-SVM (latent)	TODO	TODO	TODO	TODO	TODO	No
Elliptic envelope (raw)	TODO	TODO	TODO	TODO	TODO	No
Elliptic envelope (latent)	TODO	TODO	TODO	TODO	TODO	No

Stage 3b — Calibration-transfer ablation. We compare four deployment strategies:

1. **Absolute transfer:** deploy the simulator threshold τ_0 unchanged;

2. **Standardized bridge:** transfer z_0 and estimate local μ_k, σ_k ;
3. **Field-native offline:** estimate a percentile threshold from held-out healthy field data;
4. **Online local convergence:** initialize with z_0 and estimate τ_k^* during a short unlabeled warm-up.

This ablation directly tests whether the transferable quantity is an absolute threshold or a standardized calibration rule.

This ablation separates representation transfer from calibration transfer. The representation is transferred unchanged from simulation to field deployment; the experiment tests whether the transferable calibration quantity is the absolute threshold or its standardized position within the nominal residual distribution.

Stage 4 — Run-to-failure degradation on NASA IMS. An independently trained nominal representation is constructed from an early-life segment presumed to correspond to healthy operation for the IMS bearing family. The proposed deployment methodology is then applied unchanged: each bearing independently calibrates its operational radius during the initial nominal period and subsequently enters post-convergence monitoring. We report: (i) whether $\tau_k(t)$ converges during early life; (ii) whether residuals depart from the frozen radius before documented failure; (iii) alarm lead time; and (iv) false alarms during the accepted nominal period.

The critical distinction is between an adapting threshold and a converged operational radius. After t_{conv} , the boundary is not allowed to track the degradation trajectory.

Table 2: [PLACEHOLDER] Run-to-failure deployment outcomes on NASA IMS.

IMS stream	Failure mode	Samples to convergence	Alarm raised?	Lead time before documented failure	Healthy-period FAR
Test 1, Bearing 3	Inner race	TODO	TODO	TODO	TODO
Test 1, Bearing 4	Roller element	TODO	TODO	TODO	TODO
Test 2, Bearing 1	Outer race	TODO	TODO	TODO	TODO

Stage 5 — Edge cost. We report model size, parameter count, inference latency per 1200-sample window on CPU, local state memory, and the reduction in transmitted data when only scores and selected anomalous windows are sent rather than the continuous raw signal.

5 Results

5.1 Controlled detection

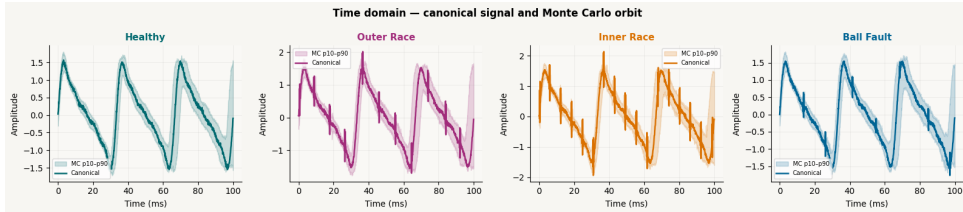


Figure 3: Canonical signals and Monte Carlo nuisance envelopes for nominal and fault regimes.

Table 3: Synthetic multi-asset detection after local operational-radius calibration.

Asset	Scenario	Nominal FAR	Post-onset DR	Samples to convergence
A	Nominal throughout	1.5%	N/A	TODO
B	Outer-race fault onset	2.0%	100%	TODO
C	Inner-race fault onset	2.0%	100%	TODO
D	Ball-fault onset	2.0%	86%	TODO

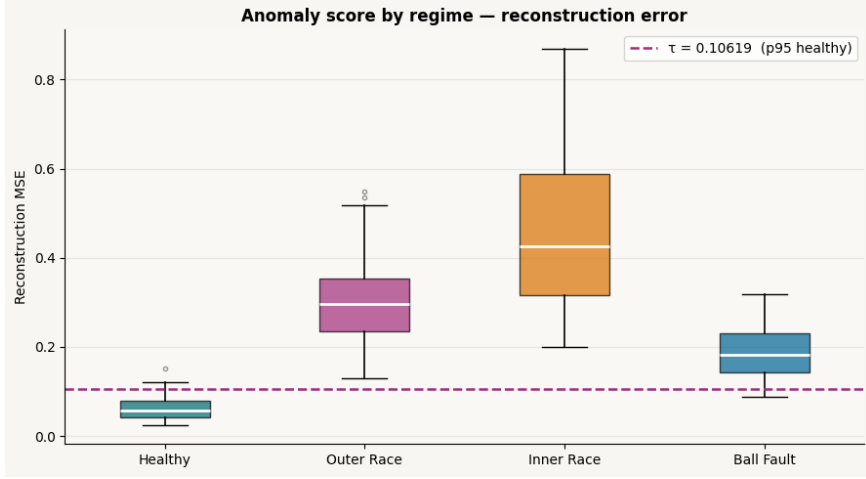


Figure 4: Reconstruction-error distributions by regime. The converged local radius defines membership in the shared nominal representation but does not modify that representation.

5.2 Operational-radius convergence

Figure 5 is the central deployment result. Stable assets should show a rapid decrease in $C_k(t)$ and convergence of $\tau_k(t)$ toward a finite asset-specific value. A degrading asset may calibrate initially, but its subsequent residual trajectory departs from the frozen τ_k^* .

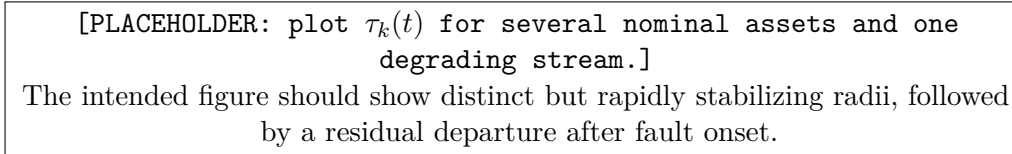


Figure 5: Per-asset operational-radius convergence. The shared representation remains fixed; only each asset’s local decision radius is identified during cold start.

5.3 Calibration transfer

On the 1797 RPM CWRU healthy stream, the reconstruction residual has $\mu_{\text{field}} = 0.2270$ and $\sigma_{\text{field}} = 0.0215$. The raw simulator threshold is expressed in incompatible amplitude units and therefore does not transfer. Re-expressing the standardized anchor in field statistics produces a decision boundary close to field-native calibration.

Table 4: Alarm rate (%) under alternative calibration-transfer strategies. For the healthy row, the value is false-alarm rate; for fault rows, it is detection rate.

Regime	Absolute transfer	Standardized bridge	Field-native	Online convergence
Healthy	100.0	11.8	5.4	11.8
Ball	100.0	99.0	98.3	99.0
Inner race	100.0	100.0	100.0	100.0
Outer race	100.0	100.0	100.0	100.0

The ablation supports the deployment claim: an absolute threshold does not survive a simulation-to-field scale change, whereas a standardized anchor combined with local statistics recovers useful behavior without field fault labels. The field-native method achieves the lowest healthy false-alarm rate because it is calibrated fully in-domain, but it requires an offline field-calibration set. Online convergence approaches that performance using only a short local warm-up.

5.4 Sensitivity and edge cost

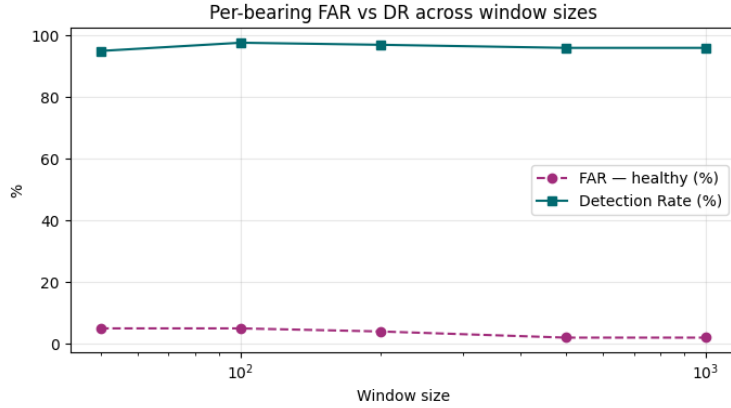


Figure 6: Sensitivity of false-alarm and detection rates to the local residual-buffer size.

Table 5: [PLACEHOLDER] Edge deployment footprint.

Metric	Value
Model parameters	847,601
Serialized model size	TODO
CPU inference latency per window	TODO
Local calibration-state memory	TODO
Raw-to-event data reduction	TODO

6 Discussion

The principal result is not that a Conv1D autoencoder can separate CWRU fault classes. That finding is already well established in the literature. The contribution is a deployment rule for heterogeneous transportation fleets: the nominal representation can be shared, while every asset independently identifies a local operational radius without model retraining.

The standardized bridge experiment shows why this separation is necessary. The numerical scale of reconstruction residuals is installation-dependent. A simulator threshold therefore fails

when treated as a physical quantity. Its standardized location within a nominal distribution is more portable, provided the deployed asset estimates its own local scale.

Operational-radius convergence also clarifies the role of adaptation. The purpose of adaptation is to identify the asset during cold start, not to follow its aging indefinitely. Once τ_k^* is established, continued movement of the anomaly-score distribution is diagnostic information. Free adaptation after convergence would instead risk absorbing a progressive fault into the nominal baseline.

For transportation practice, this architecture has several advantages. It supports day-one rollout without fault labels, removes per-asset model maintenance, isolates calibration between physical units, permits edge execution with minimal local state, and reduces continuous transfer of high-frequency raw sensor observations. The same deployment principle can be applied to bearings, traction motors, gearboxes, pumps, actuators, and spatially indexed infrastructure segments, provided the shared representation is trained for the relevant signal family.

The present evaluation remains limited. CWRU is a controlled benchmark rather than an in-service railway fleet, and fault streams are constructed rather than observed during natural field evolution. NASA IMS provides a stronger run-to-failure test but still represents laboratory rotating machinery. A later transportation-specific validation should apply the same deployment procedure to instrumented axlebox, tram, or train data with repeated observations and maintenance records. That extension is important for external validity but is not required to test the central calibration mechanism presented here.

7 Conclusion

We presented a day-one deployment methodology for nominal condition monitoring across heterogeneous asset fleets, validated in this study on public rolling-element bearing data. A single representation is trained once on simulated nominal observations and transferred unchanged to field deployment. Each operational unit then uses a short unlabeled warm-up to identify its own operational radius without modifying or retraining the shared model. The experiments show that the learned representation reuse from simulation to field data, whereas the absolute decision threshold does not. A standardized calibration anchor, re-expressed in each asset’s local residual statistics, provides the bridge between the shared representation and local deployment. The resulting primitive is label-free, independently calibrated, compatible with edge inference, and designed around fleet deployment rather than benchmark-specific classification.

A Geometric Rationale

This appendix provides a compact explanation for why a fixed nominal representation and local operational radii are compatible. It is not required for implementation.

A.1 Observation model

Let $\mathcal{X} \subset \mathbb{R}^n$ be the space of fixed-length observation windows, $\theta \in \Theta$ the latent physical state, and $\eta \in \mathcal{E}$ the nuisance variables. The observation process is

$$x = F(\theta, \eta). \quad (13)$$

Let G_{nuis} denote a physically motivated family of nuisance transformations that preserve the underlying diagnostic state. Two observations are equivalent when they differ only through such a transformation.

A.2 Compact nominal structure

Proposition 1 (Relative compactness of the nominal orbit). *If $\mathcal{E}$ is compact and $\eta \mapsto F(\theta_N, \eta)$ is uniformly bounded and equicontinuous in the sup norm, then the nominal orbit*

$$\mathcal{O}_N = \{F(\theta_N, \eta) : \eta \in \mathcal{E}\}$$

is relatively compact.

Proof. The result follows directly from the Arzelà–Ascoli theorem. Therefore $\overline{\mathcal{O}_N}$ is compact. $\square$

A representation that collapses nuisance-equivalent observations can therefore map nominal operation into a bounded structure. The shared representation approximates this structure, whereas the local radius accounts for asset-specific score scale and dispersion. Estimating τ_k^* does not deform the nominal representation; it only identifies the local acceptance boundary around it.

Proposition 2 (Factorization of diagnostic maps). *If a diagnostic map is invariant under G_{nuis} , then it is constant on nuisance orbits and factors through the quotient projection.*

This proposition motivates treating anomaly detection as membership testing in a nominal representation. It does not prescribe the autoencoder, the residual, or the calibration rule used in the main paper.

References

- [1] J. Hadamard. *Lectures on Cauchy’s Problem in Linear Partial Differential Equations*. Yale University Press, 1923.
- [2] A. N. Tikhonov and V. Y. Arsenin. *Solutions of Ill-Posed Problems*. Winston, 1977.
- [3] V. Chandola, A. Banerjee, and V. Kumar. Anomaly detection: A survey. *ACM Computing Surveys*, 41(3), Article 15, 2009.
- [4] B. Schölkopf, R. C. Williamson, A. J. Smola, J. Shawe-Taylor, and J. C. Platt. Support vector method for novelty detection. In *Advances in Neural Information Processing Systems*, 1999.
- [5] C. M. Bishop. Novelty detection and neural network validation. *IEEE Proceedings — Vision, Image and Signal Processing*, 141(4), 1994.
- [6] T. Kohonen. Self-organized formation of topologically correct feature maps. *Biological Cybernetics*, 43, 59–69, 1982.
- [7] G. E. Hinton and R. R. Salakhutdinov. Reducing the dimensionality of data with neural networks. *Science*, 313(5786), 504–507, 2006.
- [8] M. Sakurada and T. Yairi. Anomaly detection using autoencoders with nonlinear dimensionality reduction. In *Proceedings of the 2nd Workshop on Machine Learning for Sensory Data Analysis*, 2014.
- [9] M. M. Bronstein, J. Bruna, T. Cohen, and P. Veličković. Geometric deep learning: Grids, groups, graphs, geodesics, and gauges. *arXiv:2104.13478*, 2021.
- [10] A. K. S. Jardine, D. Lin, and D. Banjevic. A review on machinery diagnostics and prognostics implementing condition-based maintenance. *Mechanical Systems and Signal Processing*, 20(7), 1483–1510, 2006.

- [11] R. B. Randall and J. Antoni. Rolling element bearing diagnostics—A tutorial. *Mechanical Systems and Signal Processing*, 25(2), 485–520, 2011.
- [12] W. A. Smith and R. B. Randall. Rolling element bearing diagnostics using the Case Western Reserve University data: A benchmark study. *Mechanical Systems and Signal Processing*, 64–65, 100–131, 2015.
- [13] J. Lee, H. Qiu, G. Yu, J. Lin, and Rexnord Technical Services. IMS, University of Cincinnati. “Bearing Data Set”, NASA Ames Prognostics Data Repository. *NASA Ames Research Center*, Moffett Field, CA, 2007.
- [14] R. Zhao, R. Yan, Z. Chen, K. Mao, P. Wang, and R. X. Gao. Deep learning and its applications to machine health monitoring: A survey. *Mechanical Systems and Signal Processing*, 115, 213–237, 2019.
- [15] Y. Lei, J. Lin, Z. He, and M. J. Zuo. A review on empirical mode decomposition in fault diagnosis of rotating machinery. *Mechanical Systems and Signal Processing*, 35(1–2), 108–126, 2013.
- [16] E. Di Santi. *Self-Supervision, Manifold Saturation, and Emergent Operational Classes: Learning from Scarce Observations as Inverse Recovery of Compact Functional Structure*. Technical report, University of Colorado Boulder, 2026.
- [17] D. G. Kendall. Shape manifolds, Procrustean metrics, and complex projective spaces. *Bulletin of the London Mathematical Society*, 16(2), 81–121, 1984.
- [18] R. S. Palais. On the existence of slices for actions of non-compact Lie groups. *Annals of Mathematics*, 73(2), 295–323, 1961.
- [19] L. Li, Y. Kuang, and J. Han. Invariant anomaly detection under distribution shifts. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.